SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 10.1

Implementation of Spectral Subtraction for Speech Noise Reduction.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;

Fs=8000; t=0:1/Fs:3;
x=sin(2*pi*200*t)+0.5*sin(2*pi*400*t);   % Clean test speech-like signal
noise=0.3*randn(size(x));
xn=x+noise;                               % Noisy signal

% Estimate noise from a noise-only segment
noise_ref=noise(1:round(0.25*Fs));
N=fft(noise_ref,length(x));
Y=fft(xn);
noise_mag=mean(abs(N))*ones(size(Y));
xe=real(ifft(max(abs(Y)-noise_mag,0).*exp(1j*angle(Y))));

% Signals
figure;
subplot(311),plot(t,x),title('Clean Speech')
subplot(312),plot(t,xn),title('Noisy Speech')
subplot(313),plot(t,xe),title('Enhanced Speech')

% Spectrograms
figure;
subplot(311),spectrogram(x),title('Clean Spectrogram')
subplot(312),spectrogram(xn),title('Noisy Spectrogram')
subplot(313),spectrogram(xe),title('Enhanced Spectrogram')

% Magnitude Spectra
figure;
subplot(311),plot(abs(fft(x))),title('Clean Spectrum')
subplot(312),plot(abs(fft(xn))),title('Noisy Spectrum')
subplot(313),plot(abs(fft(xe))),title('Enhanced Spectrum')
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
% Signals
figure;
subplot(311),plot(t,x),title('Clean Speech')
subplot(312),plot(t,xn),title('Noisy Speech')
subplot(313),plot(t,xe),title('Enhanced Speech')

% Spectrograms
figure;
subplot(311),spectrogram(x),title('Clean Spectrogram')
subplot(312),spectrogram(xn),title('Noisy Spectrogram')
subplot(313),spectrogram(xe),title('Enhanced Spectrogram')

% Magnitude Spectra
figure;
subplot(311),plot(abs(fft(x))),title('Clean Spectrum')
subplot(312),plot(abs(fft(xn))),title('Noisy Spectrum')
subplot(313),plot(abs(fft(xe))),title('Enhanced Spectrum')
```

Output

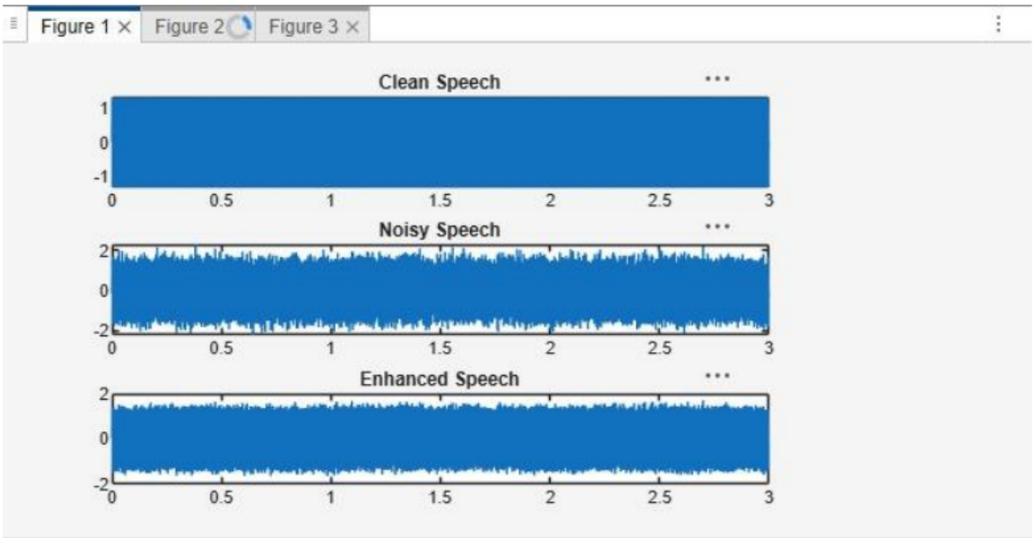


Figure 1
Figure 2
Figure 3
Clean Speech
Noisy Speech
Enhanced Speech

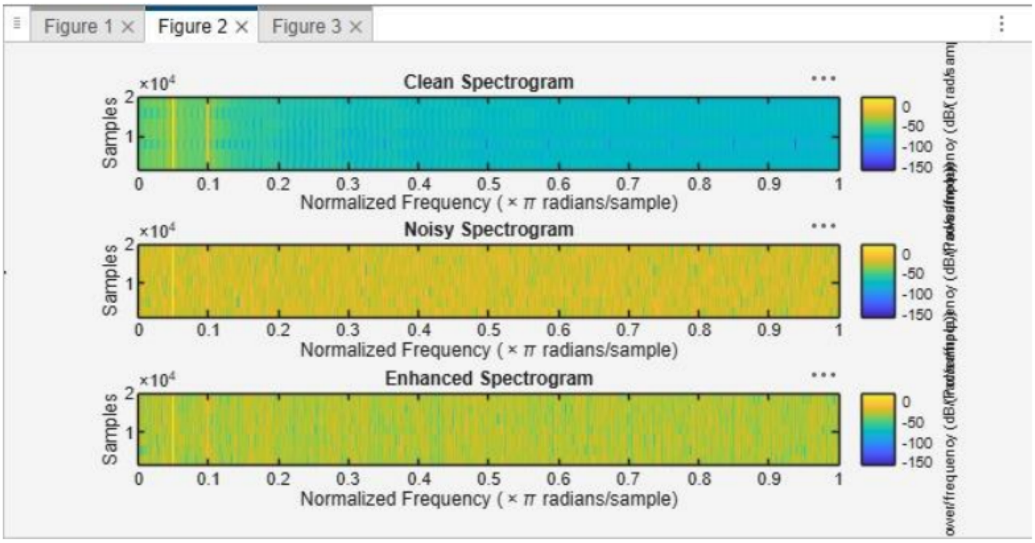
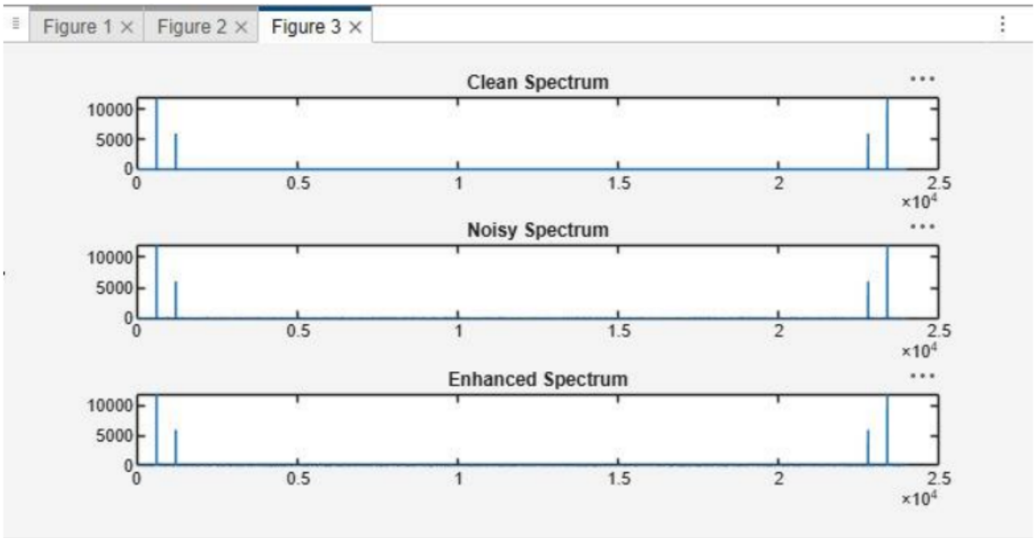


Figure 1
Figure 2
Figure 3
Clean Spectrogram
Noisy Spectrogram
Enhanced Spectrogram
Samples
Normalized Frequency (× π radians/sample)



Result

1. A noisy speech signal was successfully generated by adding noise to the clean speech signal.
2. The noise spectrum was estimated and the Spectral Subtraction algorithm was successfully implemented.
3. The magnitude spectra of the noisy and enhanced speech signals were obtained and analyzed, showing a reduction in noise components after enhancement.
4. The performance of spectral subtraction was evaluated, and the enhanced speech signal exhibited improved speech quality and intelligibility compared to the noisy speech signal.

Practical Question 1

A speech communication system is corrupted by additive white Gaussian noise. Using MATLAB, implement a Spectral Subtraction-based Speech Enhancement System with the following specifications:

- Sampling Frequency = 8 kHz
- Speech Duration = 5 seconds
- Input Speech Signal = speech.wav
- Noise Type = Additive White Gaussian Noise (AWGN)
- Input SNR = 5 dB

Tasks:

1. Generate a noisy speech signal by adding AWGN to the clean speech signal.
2. Estimate the noise spectrum and implement the Spectral Subtraction algorithm.
3. Obtain and analyze the magnitude spectra of the noisy and enhanced speech signals.
4. Evaluate the performance of the speech enhancement system using waveform and spectral analysis.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

A voice recording system is affected by background noise. Design and implement a Spectral Subtraction Speech Enhancement Technique in MATLAB with the following specifications:

- Sampling Frequency = 16 kHz
- Speech Duration = 4 seconds
- Input Speech Signal = speech.wav
- Noise Type = Babble Noise
- Input SNR = 0 dB

Tasks:

1. Generate a noisy speech signal by adding babble noise to the clean speech signal.
2. Estimate the noise spectrum from a noise-only segment and apply Spectral Subtraction.
3. Obtain and compare the magnitude spectra of the noisy and enhanced speech signals.
4. Analyze the effectiveness of noise reduction and speech quality improvement after enhancement.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To generate a noisy speech-like signal, estimate the noise spectrum, apply spectral subtraction, and compare the clean, noisy, and enhanced signals.

Student Details

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